

Curriculum Vitae - Carlos Payá

PhD candidate in Theoretical Condensed Matter Physics

Madrid, Spain · [ICMM](#) - [CSIC](#)

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Full name: Carlos Payá Herrero

Profile

I am a theoretical condensed matter physicist pursuing my PhD at the [QUDYMA](#) and [Q4Q](#) groups at [ICMM](#), [CSIC](#), in Madrid, under the supervision of [Elsa Prada](#) and [Ramón Aguado](#).

My research focuses on one-dimensional hybrid semiconductor-superconductor systems, with particular emphasis on full-shell nanowires. I combine large-scale numerical simulations with analytical modelling to address open questions spanning topological superconductivity, transport phenomena, and the phase structure of these devices.

I collaborate closely with several experimental groups (see [publications](#)) and carried out a research visit to [Karsten Flensberg's group](#) at the University of Copenhagen.

Education

2023–present · PhD in Condensed Matter Physics

Universidad Autónoma de Madrid

Tentative completion: February 2027

2021–2022 · MSc in Condensed Matter Physics

Universidad Autónoma de Madrid

MSc thesis: [Topological phase and Majorana zero modes in full-shell nanowires](#)

2017–2021 · BSc in Physics

Universidad Autónoma de Madrid

BSc thesis: [Josephson junctions between full-shell Majorana nanowires](#)

Research positions

2023–present · PhD Candidate

Instituto de Ciencia de Materiales de Madrid (ICMM), CSIC

Supervisors: Elsa Prada and Ramón Aguado · Groups: QUDYMA and Q4Q

Apr–Jul 2025 · Visiting PhD Candidate · Niels Bohr Institute, University of Copenhagen · Supervisor: Karsten Flensberg

2021–2022 · Research Assistant

Instituto de Ciencia de Materiales de Madrid (ICMM), CSIC

Supervisor: Elsa Prada

Teaching

2024–2025 · BSc Thesis Co-supervision

César Robles

Main supervisor: Elsa Prada

Quasi-Majoranas in inhomogeneous full-shell hybrid nanowires

Publications

Preprints

- [P9] **Absence of Quasi-Majorana False Positives in Full-Shell Hybrid Nanowires**
Carlos Payá, César Robles, Pablo San-Jose, Elsa Prada
[arXiv:2604.24858 \(2026\)](#)

Published papers

- [P8] **Non-Hermitian skin effect and electronic nonlocal transport**
Carlos Payá, Oliver Solow, Elsa Prada, Ramón Aguado, Karsten Flensberg
[Phys. Rev. B 113, L161405 \(2026\)](#) · [arXiv:2510.00921 \(2025\)](#)
- [P7] **Fluxoid valve effect in full-shell nanowire Josephson junctions**
Carlos Payá, F. J. Matute-Cañadas, A. Levy Yeyati, Ramón Aguado, Pablo San-Jose, Elsa Prada
[Phys. Rev. B 112, 134520 \(2025\)](#) · [arXiv:2504.16989 \(2025\)](#)
- [P6] **Josephson effect and critical currents in trivial and topological full-shell hybrid nanowires**
Carlos Payá, Ramón Aguado, Pablo San-Jose, Elsa Prada
[Physical Review B 111, 235420 \(2025\)](#) · [arXiv:2503.09756 \(2025\)](#)
- [P5] **Caroli-de Gennes-Matricon Analogs in Full-Shell Hybrid Nanowires**
M. T. Deng, Carlos Payá, Pablo San-Jose, Elsa Prada, C. M. Marcus, S. Vaitiekėnas
[Phys. Rev. Lett. 134, 206302 \(2025\)](#) · [arXiv:2501.05419 \(2025\)](#)
- [P4] **Absence of Majorana oscillations in finite-length full-shell hybrid nanowires**
Carlos Payá, Pablo San-Jose, Carlos J. Sánchez Martínez, Ramón Aguado, Elsa Prada
[Phys. Rev. B 110, 115417 \(2024\)](#) · [arXiv:2407.06330 \(2024\)](#)
- [P3] **InP/GaSb core-shell nanowires: a novel hole-based platform with strong spin-orbit coupling for full-shell hybrid devices**
Andrea Vezzosi, Carlos Payá, Paweł Wójcik, Andrea Bertoni, Guido Goldoni, Elsa Prada, Samuel D. Escribano
[SciPost Phys. 18, 069 \(2025\)](#) · [arXiv:2405.07651 \(2024\)](#)
- [P2] **Phenomenology of Majorana zero modes in full-shell hybrid nanowires**
Carlos Payá, Samuel D. Escribano, Andrea Vezzosi, Fernando Peñaranda, Ramón Aguado, Pablo San-Jose, Elsa Prada
[Phys. Rev. B 109, 115428 \(2024\)](#) · [arXiv:2312.11613 \(2023\)](#)
- [P1] **Theory of Caroli-de Gennes-Matricon analogs in full-shell hybrid nanowires**
Pablo San-Jose, Carlos Payá, C. M. Marcus, S. Vaitiekėnas, Elsa Prada
[Phys. Rev. B 107, 155423 \(2023\)](#) · [arXiv:2207.07606 \(2022\)](#)

Conference contributions

- | | | |
|-------|---|---------------|
| [C19] | International Workshop on Superconductor Semiconductor Hybrids | February 2026 |
| | | POSTER |
| | <i>Non-Hermitian Skin Effect and Electronic Nonlocal Transport</i>
Institut Néel, CNRS · Villard-de-Lans, France
Materials: Presentation PDF · Reference list | |
| [C18] | International Workshop on Superconductor Semiconductor Hybrids | February 2026 |

Realistic Modeling of 1D Hybrid Superconductors

Institut Néel, CNRS · Villard-de-Lans, France

Materials: [Presentation PDF](#) · [Reference list](#)

[C17] **QTYR25**

July 2025

CONTRIBUTED

Fluxoid valve effect in full-shell nanowire Josephson junctions

PhD and Young Scientists in Quantum Technologies Network (PYSQT) · Madrid, Spain

Materials: [Presentation PDF](#)

[C16] **Quantum Designer 2025**

July 2025

POSTER

Josephson effect in topological full-shell hybrid nanowires

Donostia International Physics Center (DIPC) · San Sebastián, Spain

Materials: [Presentation PDF](#) · [Reference list](#)

[C15] **Workshop on Superconductor-Semiconductor Hybrids**

June 2025

POSTER

Josephson effect in topological full-shell hybrid nanowires

Niels Bohr Institute, University of Copenhagen · Copenhagen, Denmark

Materials: [Presentation PDF](#) · [Reference list](#)

[C14] **GRC on Hybrid Superconductor-Semiconductor Devices**

April 2025

POSTER

Josephson effect in topological full-shell hybrid nanowires

Gordon Research Conferences · Les Diablerets, Switzerland

Materials: [Presentation PDF](#) · [Reference list](#)

[C13] **APS Global Physics Summit 2025**

March 2025

CONTRIBUTED

Josephson effect and critical currents in topological full-shell hybrid nanowires

American Physical Society · Anaheim, CA, US

Materials: [Presentation PDF](#)

[C12] **QTYR24**

July 2024

CONTRIBUTED

Phenomenology of Majorana zero modes in full-shell hybrid nanowires

PhD and Young Scientists in Quantum Technologies Network (PYSQT) · Madrid, Spain

Materials: [Presentation PDF](#)

[C11] **Quantum Designer 2024**

July 2024

POSTER

Phenomenology of Majorana zero modes in full-shell hybrid nanowires

Donostia International Physics Center (DIPC) · San Sebastián, Spain

Materials: [Presentation PDF](#)

[C10] **Quantum matter for Quantum Technologies Workshop**

May 2024

POSTER

Phenomenology of Majorana zero modes in full-shell hybrid nanowires

SPICE · Mainz, Germany

Materials: [Presentation PDF](#)

[C9] **Condensed Matter PhD Program Annual Meeting**

May 2024

CONTRIBUTED

Phenomenology of Majorana zero modes in full-shell hybrid nanowires

Facultad de Ciencias, Universidad Autónoma de Madrid (UAM) · Madrid, Spain

Materials: [Presentation PDF](#)

[C8] **European School on Superconductivity and Magnetism in Quantum Materials**

April 2024

POSTER

Phenomenology of Majorana zero modes in full-shell hybrid nanowires

SuperQmap COST action · Gandía, Spain

Materials: [Presentation PDF](#)

[C7] **Workshop on Superconductor-Semiconductor Hybrids**

March 2024

ATTENDED

Niels Bohr Institute, University of Copenhagen · Copenhagen, Denmark

[C6] **Emergence of Quantum Phases in Novel Materials**

September 2023

POSTER

Majorana zero modes in full-shell hybrid nanowires

Instituto de Ciencia de Materiales de Madrid (ICMM), CSIC · Madrid, Spain

Materials: [Presentation PDF](#)

[C5] **Bound States in Superconducting Nanodevices**

June 2023

POSTER

Theory of Caroli-de Gennes-Matricon analogs in full-shell hybrid nanowires

TopSquad and AndQC collaborations · Budapest, Hungary

Materials: [Presentation PDF](#)

[C4] **YouMat2023**

May 2023

POSTER

Theory of Caroli-de Gennes-Matricon analogs in full-shell hybrid nanowires

Instituto de Ciencia de Materiales de Madrid (ICMM), CSIC · Madrid, Spain

Materials: [Presentation PDF](#)

[C3] **QuantumMatter 2023**

May 2023

POSTER

Theory of Caroli-de Gennes-Matricon analogs in full-shell hybrid nanowires

Phantoms Foundation · Madrid, Spain

Materials: [Presentation PDF](#)

[C2] **Frontiers in Condensed Matter Physics**

July 2022

POSTER

Theory of Caroli-de Gennes-Matricon analogs in full-shell hybrid nanowires

Niels Bohr Institute, University of Copenhagen · Copenhagen, Denmark

Materials: [Presentation PDF](#)

[C1] **Emergence of Quantum Phases in Novel Materials**

September 2021

Instituto de Ciencia de Materiales de Madrid (ICMM), CSIC · Madrid, Spain

Outreach events

Since 2023, I have been a member of the Superconductivity Outreach Team at ICMM, CSIC. The group is coordinated by María José Calderón and Leni Bascones, and we run regular talks and demonstrations for high-school students and the general public.		
[09]	Feria Madrid es Ciencia 2026 (Madrid Science Fair)	March 2026 WORKSHOPPER
Comunidad de Madrid · Madrid, Spain		
[08]	European Researchers' Night 2025	September 2025 WORKSHOPPER
CSIC · Madrid, Spain		
[07]	Feria Madrid es Ciencia 2025 (Madrid Science Fair)	March 2025 WORKSHOPPER
Comunidad de Madrid · Madrid, Spain		
[06]	European Researchers' Night 2024	September 2024 WORKSHOPPER
CSIC · Madrid, Spain		
[05]	Feria Madrid es Ciencia 2024 (Madrid Science Fair)	March 2024 WORKSHOPPER
Comunidad de Madrid · Madrid, Spain		
[04]	Semana de la Ciencia (Science Week)	November 2023 WORKSHOPPER
CSIC · Madrid, Spain		
[03]	European Researchers' Night 2023	September 2023 WORKSHOPPER
Instituto de Ciencia de Materiales de Madrid (ICMM), CSIC · Madrid, Spain		
[02]	Ciencia en la Calle	June 2023 WORKSHOPPER
Casa de la Ciencia · Ciudad Real, Spain		
[01]	European Researchers' Nigth 2022	September 2022 LOGISTICS
Instituto de Ciencia de Materiales de Madrid (ICMM), CSIC · Madrid, Spain		

Funding

[F3]	PhD Fellowship	2023-2027
AEI · PRE2022-101362		
ICMM (CSIC)		
[F2]	Correlations, Superconductivity and Topology in Quantum Materials and Technologies	2022-2025
AEI · PID2021-125343NB-I00		

ICMM (CSIC)

PI: Ramón Aguado and Elena Bascones

[F1] **Topology and Correlations in Quantum Materials and Solid State Quantum Technologies** 2021-2022

AEI · PGC2018-097018-B-I00

ICMM (CSIC)

PI: María José Calderón and Ramón Aguado

Awards

[A4] **Max Mazín Award** 2017-2021

Max Mazín Foundation and CEIM Foundation

Awarded each year of my undergraduate studies, 5th to 8th editions

[A3] **GEFES Research Award for Students** 2020

Condensed Matter Physics Division, Spanish Royal Society of Physics (GEFES-RSEF)

Josephson Junctions in Full-shell Majorana Nanowires.

[A2] **Excellence Fellowship** 2018

Comunidad de Madrid

[A1] **Premio Extraordinario de Bachillerato** 2017

Comunidad de Madrid

Top 10 academic records of the region

Skills

Research methods: Analytical modelling, numerical simulation of quantum transport and hybrid superconducting devices, data analysis and visualization.

Programming: Python, Julia, Matlab, scientific scripting and reproducible computational workflows.

Scientific tools: Git/GitHub, LaTeX, Jupyter, Linux-based HPC environments.

Languages: Spanish (native), English (C1), French (C1).

Printable CV is ready.